Part I – General Review

Exercise 1

Drakeley et al (2005) described a malaria study in northeast Tanzania. In this study, the authors conducted a stratified survey by 6 transect (Rombo; South Pare; North Pare; West Usambara 1, 2, and 3), 4 villages per transect (one village at high altitude, >1200m; two villages at intermediate altitude, 600m-1200m and one village at low altitude, <600m; and age class (0-4, 5-14 and 15-45 years old).

a) How many strata are defined by study design? Calculate the sample size per stratum.
b) Discuss the reason why the authors used transect, altitude and age class as defining stratum.
c) What is the sampling distributions associated with the gender?
d) The authors claimed that the gender distribution was matched across villages. What does it mean matched distribution?
e) Conduct an appropriate statistical test to assess the validity of matching gender distribution across villages. Draw your conclusions.
f) The authors claimed that the survey intended to sample 32%, 32% and 36% of individuals from the age classes 0-4, 5-14 and 15-45 years old, respectively. Perform a convenient statistical test to assess that claim.

Data file:

data_tanzania.csv

Reference:

Drakeley CJ, Carneiro I, Reyburn H, Malima R, Lusingu JP, Cox J, Theander TG, Nkya WM, Lemnge MM, Riley EM. Altitude-dependent and -independent variations in Plasmodium falciparum prevalence in northeastern Tanzania. J Infect Dis. 2005 May 15;191(10):1589-98. doi: 10.1086/429669.

Part II – Applications to Diagnostics

Exercise 1

Aneityum is a small island of the Vanuatu archipelago located in the Pacific ocean. The island, also called Mystery island, has a total of 700 inhabitants. In the last decade, there was a local effort to eliminate malaria. Imagine you conducted a random survey of 50 individuals with the objective of estimating the total number of individuals who are currently infected in the population. After conducting the survey, you found two infected individuals.

a) What is the sampling model (or distribution) for these data?

b) Estimate the total number of individuals who are currently infected in the population using the maximum likelihood method and the method of moments.

Exercise 2:

Malaria remains a significant public health problem in Thailand, especially, in the northwestern province of Tak along the Thai–Myanmar border. As part of the national malaria control and surveillance program, three cross-sectional surveys were conducted in the village of Mae Salid Noi between March 2013 and January 2014. The surveys were conducted in both community and hospital/malaria clinic separately. The objective of the study was to assess the current malaria prevalence using different methods to detect malaria infections. The table below shows the frequency of individuals with malaria infection detected by molecular (qPCR) or microscopy testing in the community and hospital/malaria clinic surveys.

		Microscopy result	
Survey	QPCR result	Positive	Negative
Community	Positive	3	38
	Negative	0	1306
hospital/malaria clinic	Positive	26	6
	Negative	0	265

a) Assume the molecular testing as your gold standard and estimate the sensitivity and specificity of microscopy testing in community and hospital/malaria clinic separately. Calculate the respective 95% confidence intervals.

b) Is the Wald's confidence interval advisable to use for estimating sensitivity and specificity? Explain you reasoning.
c) Discuss your results by considering in which situation, if any, microscopy is a recommended diagnostic tool for malaria detection.

Reference:

Baum E, Sattabongkot J, Sirichaisinthop J, Kiattibutr K, Jain A, Taghavian O, Lee MC, Huw Davies D, Cui L, Felgner PL, Yan G. Common asymptomatic and submicroscopic malaria infections in Western Thailand revealed in longitudinal molecular and serological studies: a challenge to malaria elimination. Malar J. 2016;15:333.

Exercise 3

Serological (or antibody) testing is a general diagnostic technique for infectious disease based on the principle of detecting antibodies against specific proteins of a given pathogen, such as the human immuno-defficiency virus (HIV), the causative infection agent of acquired immunodefficiency syndrome (AIDS). Unfortunately, serological testing can provide indeterminate results. To understand the determinants of these results in HIV testing, a population-based study was conducted in the Rakai District of Uganda between July 2013 and May 2018. The table below shows the frequency of individuals with positive, indeterminate, and negative test results by sex.

	Result of HIV antibody testing		
Sex	Positive	Indeterminate	Negative
Male	2966	455	22085
Female	4706	1035	23222

a) Estimate the proportion of HIV indeterminate results by sex.
b) Calculate the corresponding 95% confidence region for the proportion of positive and negative results using the Bonferroni correction method.
c) Perform a statistical test to understand whether the sex of the individuals influence the results of HIV antibody testing.

Reference:

Hunt JH, Mwinnyaa G, Patel EU, Grabowski MK, Kagaayi J, Gray RH, Ssekasanvu J, Wawer MJ, Kigozi G, Chang LW, Kalibbala S, Nakalanzi M, Ndyanabo A, Quinn TC, Serwadda D, Reynolds SJ, Galiwango RM, Laeyendecker O. Longitudinal patterns in indeterminate HIV rapid antibody test results: a population-based, prospective cohort study. Microbiol Spectr. 2024;12(2):e0325323.

Part III – Applications to Medicine

Exercise 1